Salon '26

AI From Insight to Impact That Matters

The Center for Health Outcomes and Informatics Research

Platform	**Google Colab**
Language	**Python 3**

Interacting with

Large Language Models in Python

From foundational concepts to working code

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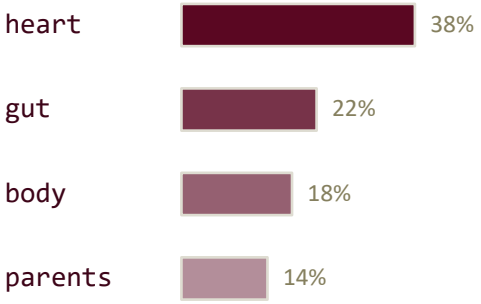
What is a Large Language Model?

A probabilistic model of natural language, trained to predict the next token given all preceding context.

Input: `"Listen to your ___ "`

Model output P(next token):



Trained on trillions of tokens → the model learns grammar, facts, code, and reasoning as emergent side-effects.

Three Levels of Working with LLMs

Ordered by required expertise and computational resources.

	I Prompt Engineering *No parameter changes*	**II Fine-Tuning** *Adapt ≥1 weight*	**III Pre-training** *Train from scratch*
Data needed	**None**	1K–100K examples	Billions of tokens
Compute	**API calls only**	GPUs (hours–days)	$M+ infrastructure
Expertise	**Basic Python**	ML engineering	ML research team
Example	summarize customer reviews	Fine-tune a model to classify customer support tickets or detect sentiment	Train a new language model on **company data for general-purpose use**

Level I — Prompt Engineering: Access Methods

Two ways to interact with an LLM out-of-the-box.

A — Chat Interface

e.g. ChatGPT, Claude.ai

+ No code required
+ Immediately accessible
+ Free to use

− Not automatable
− No parameter control
− Cannot integrate into applications

Suitable for exploration and ideation.

B — Programmatic API

e.g. Groq API ← this session

+ Full parameter control
+ Automatable and scalable
+ Integrable into products

− Requires Python
− API key management

Suitable for research, automation, products.

Level II — Model Fine-Tuning

Adjusting at least 1 (internal) model parameter (but not all) for a particular task.

Figure 1. Fine-tuning a pre-trained LLM for task-specific performance.

Level III — Build Your Own LLM

Come up with all model parameters from scratch.

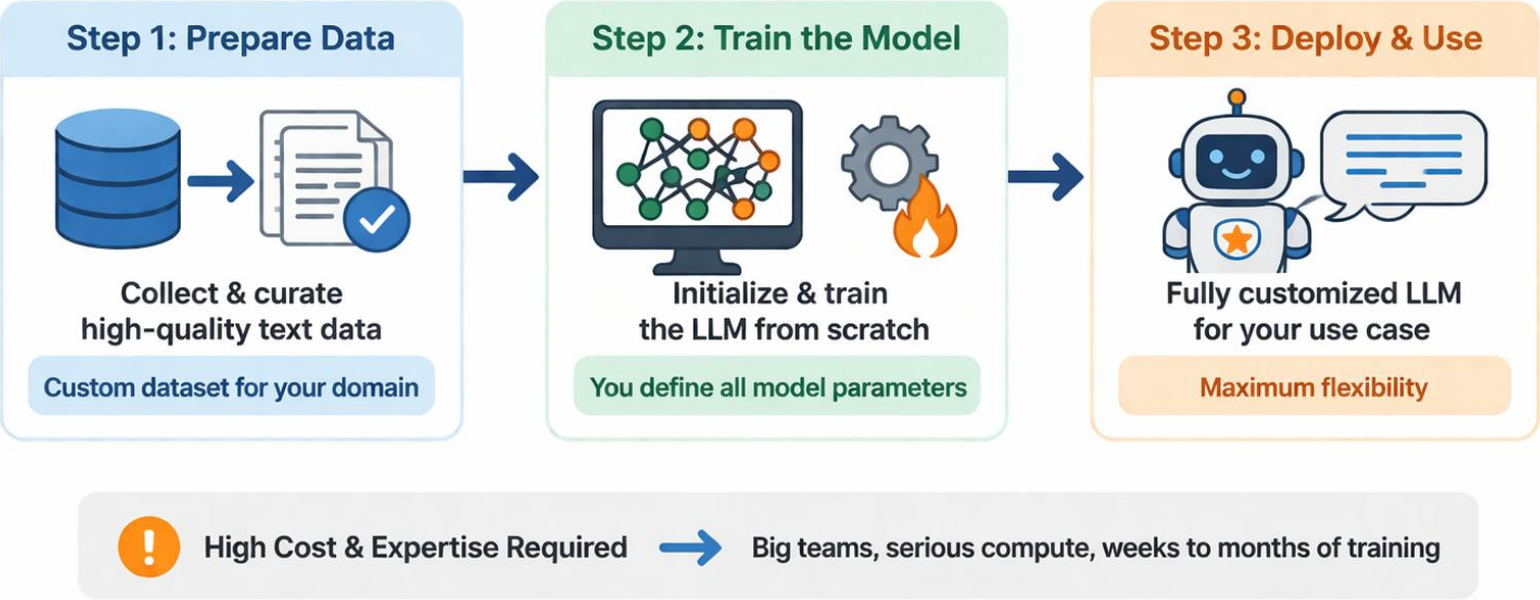


Figure 2: Training and deploying an LLM from scratch.

How We Use LLMs in Practice

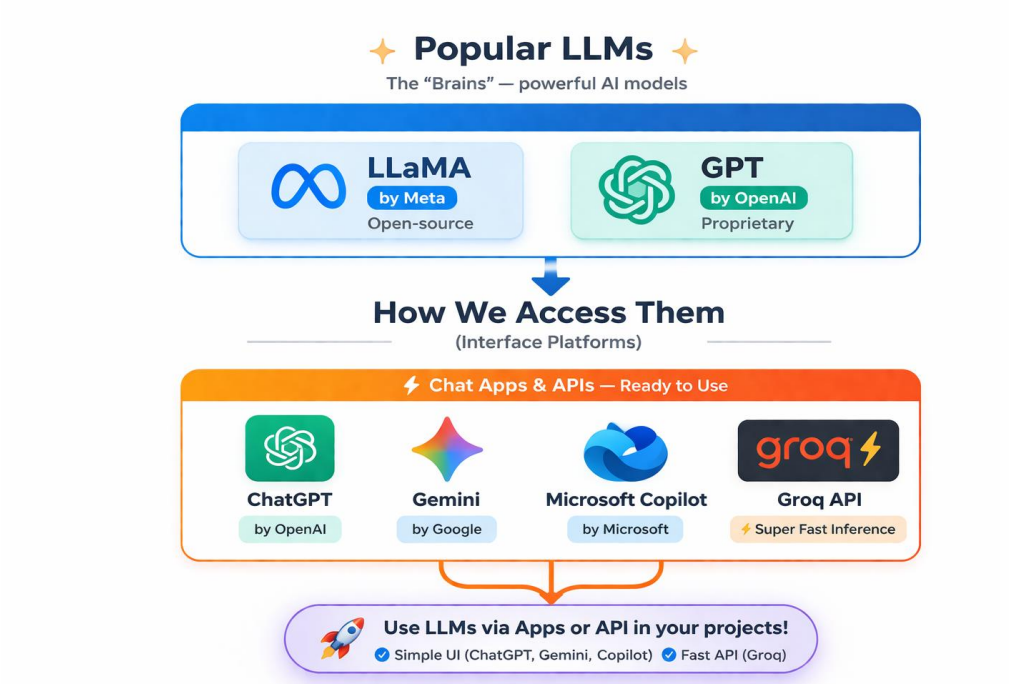


Figure 3: LLMs + Apps/APIs = Real AI use

Why Groq?

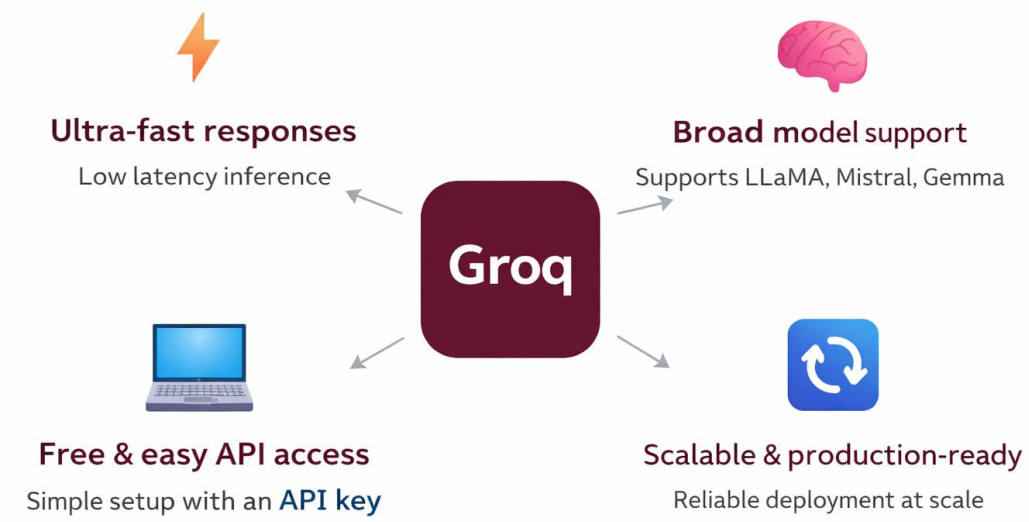


Figure 4: Groq provides fast and easy access to LLMs.

How We Use LLMs with "*Groq*" API:

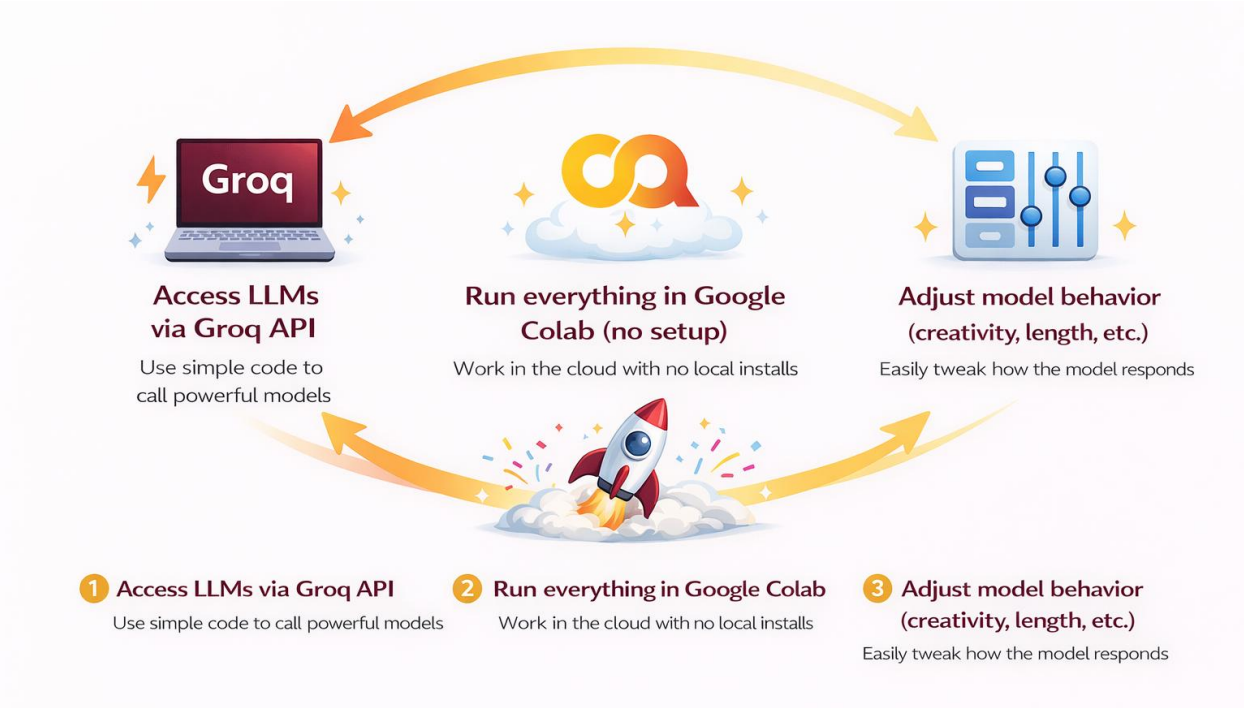


Figure 5: Simple workflow to access and use LLMs with Groq and Google Colab.

Thank You!

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